Asia Pacific Management Review 31 (2026) 100450

HOSTED BY

Contents lists available at ScienceDirect

Asia Pacific Management Review

journal homepage: www.elsevier.com/locate/apmrv

Uncovering customer insights from online hotel reviews in Vietnam: A bilingual NLP study using sentiment analysis and topic modelling

Thi-Kim-Hien Le [a,b,*], Dang Thi Kieu Anh [a,b], Ngo Thi Thanh Huyen [a,b], Le Nguyen Hoang Kim [a,b], Huynh Le Dong Quan [a,b], Pham Tran Thu Huong [a,b]

[a] University of Economics and Law, Ho Chi Minh City, Vietnam
[b] Vietnam National University, Ho Chi Minh City, Vietnam

ARTICLE INFO

Article history:
Received 5 December 2025
Received in revised form
6 May 2026
Accepted 14 August 2026
Available online 25 August 2026

Keywords:
Customer review analysis
Natural language processing (NLP)
Sentiment analysis
Topic modeling
Hospitality industry

ABSTRACT

Online customer reviews are a critical source of insight for evaluating hotel service quality in the current digital era. This study analyzes over 150,000 English and Vietnamese reviews from TripAdvisor, Booking.com, and Agoda, using a multilingual NLP framework. DistilBERT and PhoBERT were employed for sentiment classification, while BERTopic extracted latent themes. The datasets show a strong positivity bias. A combined RandomOverSampler and K-fold strategy was used to address class imbalance and improve sentiment prediction. The results indicate that BERTopic outperforms LDA in coherence and interpretability, producing fine-grained topics related to staff service, cleanliness, amenities, food quality, and value for money. Cross-language comparisons reveal distinct patterns in how English and Vietnamese guests articulate negative sentiment, emphasizing the need for culturally sensitive analysis. The study provides a scalable method for multilingual hotel review analytics and offers actionable insights to further enhance service in Vietnam's hospitality industry.

© 2026 The Authors. Published by Elsevier B.V. on behalf of College of Management, National Cheng Kung University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

Vietnam has become one of Southeast Asia's fastest-growing tourism destinations, with international search demand for accommodations increasing by more than 75% from 2023 to 2025 (Google Destination Insights). As competition intensifies, online reviews have become a decisive source of information that influences traveler perceptions and booking decisions (Jayathilaka et al., 2020; Kukreti et al., 2024). Industry statistics indicate that 81% of travelers consult online reviews before choosing accommodations (Hotel Tech Report, 2026), and approximately 52% avoid hotels with no reviews (Tripadvisor, 2019). Consequently, user-generated content on platforms such as TripAdvisor, Booking.com, and Agoda provides valuable, experience-based insights that can inform hotel service improvement and strategic decision-making (Luo et al., 2021; Marcolin et al., 2021)

Despite the growing adoption of Natural Language Processing (NLP) in hospitality research, several limitations remain. Multilingual reviews, reflecting the linguistic and cultural heterogeneity of global tourism, are often processed using monolingual or general multilingual models that degrade in performance on mixed-language datasets (Alaei et al., 2017; Ngo & Valerievich, 2024). In addition, many studies rely on a single review platform, resulting in biased representations and limited generalizability (Ding et al., 2023; Le & Huh, 2021). Traditional topic modeling techniques such as Latent Dirichlet Allocation (LDA) are further limited by the Bag-of-Words assumption and the requirement to predefine the number of topics, making them less effective at capturing the contextual and semantic richness of hotel reviews (Blei et al., 2003; O'Callaghan et al., 2015).

To address these gaps, this study analyzes a multilingual corpus of more than 150,000 English and Vietnamese hotel reviews collected from TripAdvisor, Booking.com, and Agoda. Rather than merely integrating existing NLP components, this study aims to investigate how multilingual and imbalanced data characteristics

This article is part of a special issue entitled: SI: 26th APMC published in Asia Pacific Management Review.

* Corresponding author. University of Economics and Law, VNU-HCM, Ho Chi Minh city, Viet Nam.
E-mail addresses: hienltk@uel.edu.vn (T.-K.-H. Le), anhdtk22406c@st.uel.edu.vn (D.T.K. Anh), huyenntt22406c@st.uel.edu.vn (N.T.T. Huyen), kimlnh22406c@st.uel.edu.vn (L.N.H. Kim), quanhld22416c@st.uel.edu.vn (H.L.D. Quan), huongptt22406c@st.uel.edu.vn (P.T.T. Huong).

https://doi.org/10.1016/j.apmrv.2026.100450
1029-3132/© 2026 The Authors. Published by Elsevier B.V. on behalf of College of Management, National Cheng Kung University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

influence model behavior and the extraction of customer insights. The research implements a unified transformer-based pipeline that employs DistilBERT for English sentiment analysis (Devlin et al., 2018; Sanh et al., 2019) and PhoBERT for Vietnamese (Nguyen & Nguyen, 2020), enabling more accurate modeling of linguistic nuances in both languages. Given the substantial sentiment imbalance commonly observed in hotel reviews, where positive labels dominate across datasets, the study treats class imbalance as a practical challenge in real-world data. It applies appropriate handling strategies to improve model performance, rather than conducting a dedicated analysis of its theoretical impact. An integrated RandomOverSampler and K-fold ensemble approach is used to enhance model robustness and improve classification performance for neutral and negative sentiments, outperforming conventional techniques such as SMOTE (Singgalen, 2024).

For topic extraction, the study applies BERTopic (Grootendorst, 2022), not only as a modeling choice but also to explore whether embedding-based topic models can better capture fine-grained service attributes in multilingual settings than traditional probabilistic approaches. This approach addresses well-known limitations of LDA, enabling the discovery of more coherent, fine-grained topics and yielding superior interpretability for hospitality analytics (Abuzayed & Al-Khalifa, 2021; Egger & Yu, 2022). By training BERTopic separately for English and Vietnamese reviews and validating topics using coherence measures such as C_V and NPMI, the study ensures both semantic precision and cross-linguistic rigor in topic discovery.

The empirical findings reveal several noteworthy insights. The datasets exhibit strong positivity bias across all platforms, echoing patterns reported in previous tourism sentiment studies (Mehraliyev et al., 2022). More importantly, the results suggest that such an imbalance is associated with challenges in model performance, particularly in predicting minority sentiment classes. Clear linguistic differences emerge in how English and Vietnamese reviewers express dissatisfaction, revealing a systematic bilingual asymmetry in sentiment expression, in which users differ in both the linguistic intensity and structure of negative feedback. This finding highlights the importance of culturally and linguistically aware sentiment analysis in multilingual environments. Topic modeling results yield 18 Vietnamese topics and 41 English topics, capturing key service dimensions such as staff behavior, room quality, cleanliness, amenities, food services, and perceived value. The results further show that embedding-based topic modeling enables the discovery of more fine-grained and semantically rich service attributes that are not easily captured by traditional models such as LDA, particularly in multilingual datasets. These findings align with and extend prior research on customer experience in Vietnam's hotel sector (Nguyen et al., 2023; Nguyen & Nguyen, 2023). The higher coherence achieved by BERTopic compared with LDA further demonstrates the suitability of embedding-based topic models for multilingual hospitality analytics.

By integrating multilingual transformer-based sentiment classification with advanced semantic topic modeling, this study provides a scalable and methodologically sound framework for analyzing large-scale customer reviews in the hospitality domain. The study offers empirical observations on cross-linguistic differences in sentiment expression and highlights class imbalance as a practical challenge in real-world sentiment analysis. In addition, it demonstrates the advantage of embedding-based topic modeling in capturing fine-grained service attributes in multilingual datasets. The contributions advance existing NLP applications in tourism research and offer actionable insights to support service enhancement, quality management, and data-driven decision-making in Vietnam's rapidly developing hotel industry.

2. Literature review

Research on sentiment analysis and topic modeling in tourism and hospitality has grown substantially over the past decade, reflecting the growing importance of user-generated content in shaping traveler decision-making (Tan et al., 2023). Early studies primarily employed lexicon-based or traditional machine-learning approaches to classify polarity in online reviews and extract key themes from textual data (Mahdikhani & Meena, 2024; Saidi et al., 2022). Lexicon-driven techniques, while useful for domain-independent applications, often perform inconsistently when faced with implicit sentiment or context-dependent expressions (Alaei et al., 2017; Catelli et al., 2022). Supervised machine-learning models generally outperform lexicon-based methods but remain limited in multilingual environments and struggle to capture nuanced emotional cues (Alaei et al., 2017; Kausar et al., 2019; Kim et al., 2017; Rahman et al., 2023). These constraints highlighted the need for more sophisticated methods to handle complex linguistic patterns in tourism discourse.

Topic modeling has also been widely used to explore hospitality-related themes (Abdelrazek et al., 2023). Structural topic modeling and LDA-based studies have mapped broad thematic landscapes across platforms such as Airbnb or TripAdvisor (Ding et al., 2023), while cross-platform analyses have broadened the understanding of traveler concerns across geographic contexts. However, LDA and its variants rely on Bag-of-Words assumptions and fixed numbers of topics, limiting their ability to detect context-specific and semantically rich patterns (Blei et al., 2003; Jelodar et al., 2018). In crisis-related research, topic modeling has been used to examine sentiment and communication patterns during events such as the COVID-19 pandemic, yet these studies often suffer from narrow datasets and limited topical granularity (Gulati et al., 2022; Tang et al., 2024). The broader literature consistently confirms the value of sentiment and topic modeling but also highlights persistent methodological gaps in multilingual processing, semantic depth, and the representativeness of data sources.

More recent studies have integrated deep-learning models to enhance sentiment classification performance in tourism (Atandoh et al., 2023; Mahendhiran & Kannimuthu, 2018). Nguyen and Nguyen (2023) applied VADER to 20,000 TripAdvisor reviews of Vietnamese hotels, identifying high satisfaction with location but notable dissatisfaction with room services. Le and Huh (2021) used an RNN–LSTM architecture with CBOW embeddings to extract sentiment-bearing features from Vietnamese e-commerce reviews, revealing key dimensions such as quality, price, and design. These works illustrate increased sophistication in model design but still rely on monolingual datasets and lack cross-linguistic validation.

Advances in transformer-based models have further improved sentiment analysis performance (Ngo et al., 2022). For Vietnamese, fine-tuning PhoBERT has produced strong results, with Ngo et al. (2024) achieving 92.74% accuracy in classifying mobile phone reviews. Similarly, Kumar and Sachdeva (2021) showed that combining review titles and content, along with hybrid architectures such as Ensemble BiLSTM + CNN, significantly improves classification performance, achieving an accuracy of 96% and an F1-score of 95%. These studies highlight the potential of Vietnamese-language transformers but remain confined to single-domain or single-platform datasets.

In parallel, multilingual or cross-domain applications in tourism remain sparse (Azhar & Khodra, 2021). While some studies have explored multilingual settings, most models degrade

sharply when processing multiple languages simultaneously, especially when language-specific semantics are critical (Gunadi & Setiawan, 2024). Multilingual tourism analytics also face challenges due to data fragmentation, as most existing research relies on individual platforms such as Scopus or TripAdvisor, leading to incomplete representations of traveler experiences (Ding et al., 2023; Le & Huh, 2021; Ruiz & Fernández Avilés, 2023). Consequently, there remains a need for comprehensive, cross-platform, and cross-linguistic analyses to capture the full breadth of customer feedback in emerging markets such as Vietnam.

Recent work in topic modeling has also shifted toward embedding-based approaches, with BERTopic showing improvements in flexibility and semantic coherence (Alcoforado et al., 2022; Egger & Yu, 2022; Grootendorst, 2022). Unlike LDA, BERTopic incorporates contextual embeddings and density-based clustering, allowing topics to emerge organically rather than through predefined specifications. Studies using BERTopic in hospitality or social media analytics have reported more interpretable and fine-grained themes (Abuzayed & Al-Khalifa, 2021; George & Sumathy, 2023; Gokcimen & Das, 2024), yet its application in multilingual tourism datasets, particularly in developing markets, remains limited.

Taken together, prior studies confirm the importance of sentiment and topic modeling for understanding traveler perceptions but reveal notable gaps: limited multilingual capability, reliance on single-platform datasets, dependence on conventional topic models, and insufficient exploration of emerging tourism markets such as Vietnam. Building on these insights, the present study proposes a unified, transformer-based framework for sentiment analysis and embedding-driven topic modeling across English and Vietnamese hotel reviews sourced from multiple platforms. This approach addresses the methodological constraints found in earlier research and contributes to a more comprehensive and context-aware understanding of customer experience in Vietnam's hospitality sector (Trung-Thanh et al., 2023).

3. Research models

3.1. Dataset cleaning, preprocessing, and splitting

To ensure that the analytical models accurately capture linguistic, cultural, and platform-specific characteristics, all collected reviews underwent a multi-stage cleaning, preprocessing, and splitting procedure. Fig. 1 illustrates the overall workflow, which consists of language identification, removal of non-target scripts, text normalization, and dataset separation by platform and language. This structured process ensures that English and Vietnamese reviews are handled with appropriate language-specific tools, thereby improving model precision and representational quality.

The dataset was sourced from Agoda, Booking.com, and TripAdvisor. Because Agoda contains reviews in numerous languages, multiple passes of language detection were performed using the langdetect library. In the first pass, reviews written in logographic or abugida scripts (e.g., Japanese, Chinese, Thai) were removed to reduce irrelevant linguistic noise. Subsequent passes filtered out additional non-target languages such as French, Portuguese, and German. Only English and Vietnamese reviews were retained, leading to 59,795 useable Agoda entries. Booking.com data, collected via Apify, was pre-filtered to include only English and Vietnamese text; thus, a single-language detection step was sufficient to finalize the separation. For TripAdvisor, language detection revealed that fewer than 50 of more than 70,000 reviews were written in Vietnamese; these were excluded to avoid a severe minority-class imbalance, leaving only English reviews for analysis.

Table 1 summarizes the raw and filtered data volumes after language-based splitting across the three platforms. The table shows the substantial dominance of English reviews on TripAdvisor and the relatively more balanced English–Vietnamese distribution on Agoda and Booking.com. This distributional insight matters for downstream modeling because it reflects differing user demographics across platforms.

Following language separation, a unified preprocessing pipeline was used to standardize both English and Vietnamese reviews. Fig. 2 presents the preprocessing workflow, which includes text normalization, removal of extraneous characters, handling of emojis, translation of pictograms, segmentation, and stopword removal. For English texts, emojis were converted into descriptive tokens using the emoji library, while pictograms from Chinese, Japanese, or Korean scripts were identified using regular expressions and translated via DeepL to preserve semantic information. Abbreviations were expanded using NLTK, and the text was normalized through a series of operations, including punctuation standardization, URL removal, repeated character filtering, and conversion to lowercase. Stopwords were removed, and word segmentation was performed using the UITws_v1 model to ensure token-level consistency.

Vietnamese texts underwent a similar but language-specific pipeline. Because Vietnamese includes tonal markers and compound words that cannot be naively segmented using whitespace, pictograms were removed directly, embedded English tokens were translated into Vietnamese, and invalid or very short reviews were discarded. Word segmentation was then performed using UITws_v1 to ensure compatibility with PhoBERT and BERTopic, which rely on consistent token structures.

To prepare the data for sentiment analysis, each review was labeled according to the numerical rating provided on the original platform. Table 2 outlines the sentiment classification scheme. For Agoda and Booking.com, which use a 1–10 rating scale, reviews with scores ≤ 5 were labeled negative, scores between 5 and 8 were labeled neutral, and scores ≥ 8 were labeled positive. For TripAdvisor, which uses a 1–5-star system, scores ≤ 2 were labeled negative, scores between 2 and 4 were labeled neutral, and scores ≥ 4 were labeled positive. This conversion ensures cross-platform comparability while preserving the meaning of the original rating scales.

To further optimize topic modeling results, an additional stopword processing step was applied using spaCy's syntactic dependency analysis. This step retains contextually important words, specifically nouns, verbs, adjectives, and adverbs, while removing syntactically peripheral tokens, thereby improving topic cluster quality and keyword distinctiveness. Finally, additional preprocessing steps were performed to prepare the data for topic modeling using BERTopic. Stopwords in both languages were removed using comprehensive lists, lemmatization was performed to reduce words to their base forms, and punctuation and special characters were removed to prevent token fragmentation. Proper nouns and location-specific entities were standardized (e.g., multi-word locations connected with underscores) to maintain semantic coherence during clustering. These refinements produce a clean, language-consistent corpus suitable for transformer-based sentiment classification and embedding-driven topic extraction. In Tables 2 and X denotes the review score (rating) assigned to each review by users on the respective platforms.

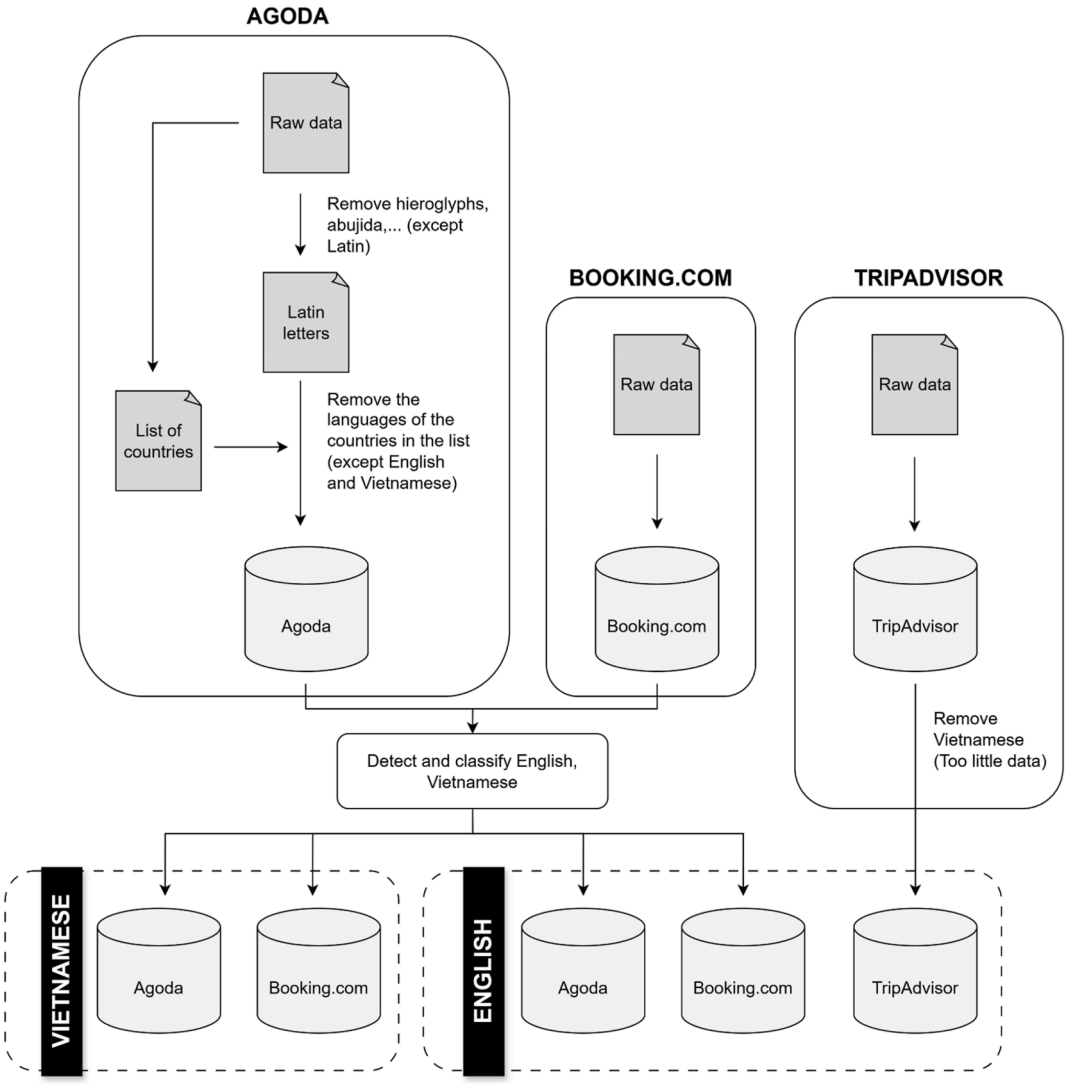


Fig. 1. Dataset splitting workflow Source: Authors.

Table 1
Summary of data volume.

		Agoda	Booking	TripAdvisor
Raw Data Volume		99.815	41.209	70.345
After Data Splitting	**Vietnamese**	16.861	2.714	
	English	39.053	38.117	70.301

3.2. Sentiment analysis

Numerous language models have been developed based on BERT (Bidirectional Encoder Representations from Transformers), one of the most advanced architectures in natural language processing (Devlin et al., 2018; Jurafsky & Martin, 2024). BERT is built on the Transformer architecture (Vaswani et al., 2017), enabling parallel word processing and the extraction of contextual meaning across entire sentences. Its bidirectional encoder captures word dependencies in depth and can be adapted to various downstream tasks. Training relies on two core objectives: the Masked Language Model (MLM), which "randomly masks certain words and requires the model to predict them based on the remaining context" (Wettig et al., 2023), and Next Sentence Prediction (NSP), a binary task that differentiates between "IsNext" and "NotNext" sentences using [CLS] and [SEP] tokens. Building on these mechanisms, BERT has inspired numerous variants fine-tuned for multiple languages and domains, establishing a strong foundation for later advances in natural language processing.

3.2.1. DistilBERT

DistilBERT is a compressed variant of BERT developed through knowledge distillation, retaining approximately 97% of BERT's language understanding performance while reducing model size by 40% and improving inference speed (Sanh et al., 2019). Rather than processing input through BERT's full 12-layer encoder, DistilBERT uses 6 layers, making it substantially more efficient without sacrificing representational quality for sentence-level tasks.

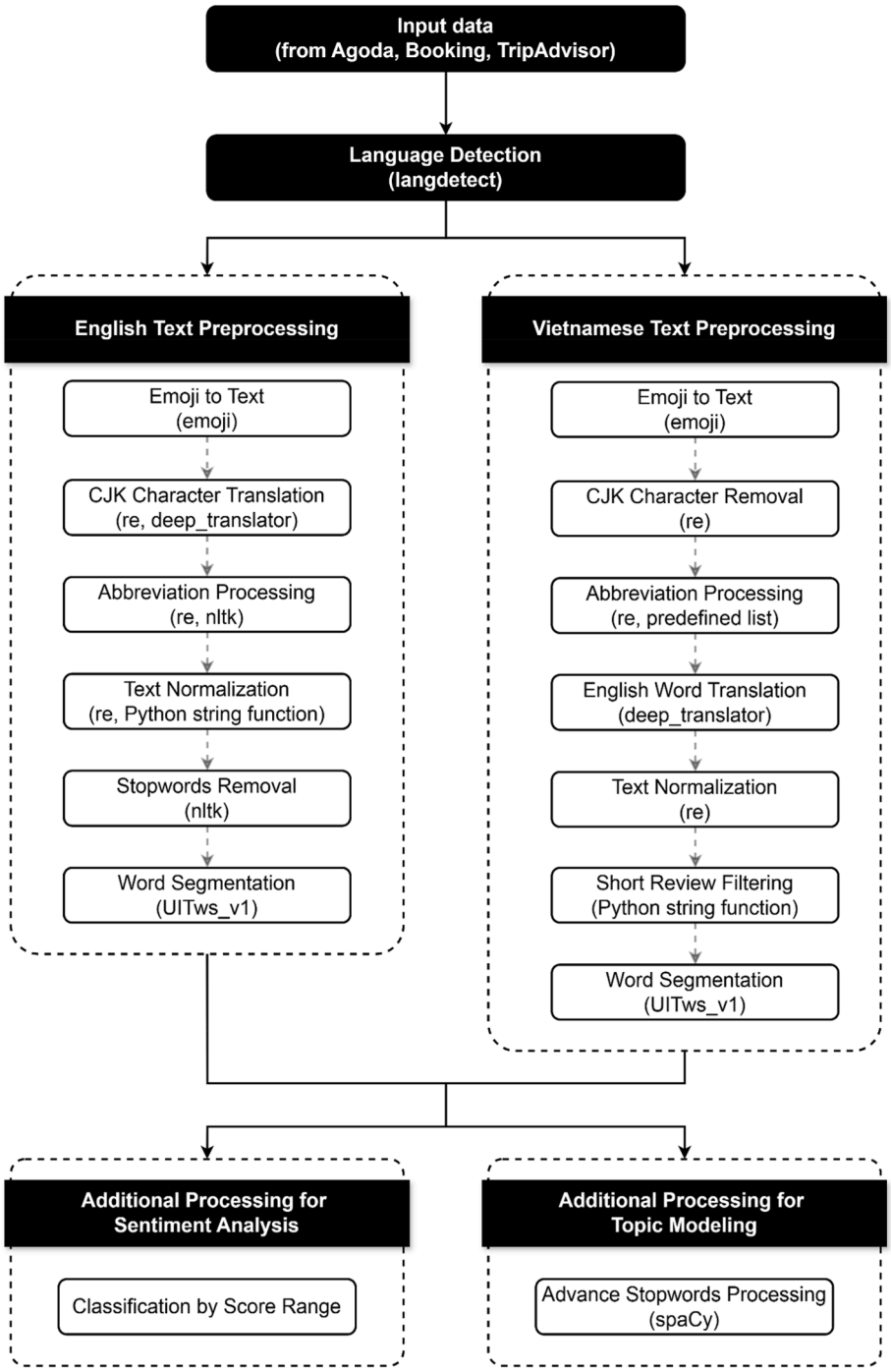


Fig. 2. Dataset preprocessing workflow.
Source: Authors

Table 2
Sentiment classification by score range.

Platform	Score Range	Negative	Neutral	Positive
Agoda	1 - 10	$X \leq 5$	$5 < X < 8$	$X \geq 8$
Booking	1 - 10	$X \leq 5$	$5 < X < 8$	$X \geq 8$
TripAdvisor	1 - 5	$X \leq 2$	$2 < X < 4$	$X \geq 4$

DistilBERT was chosen over BERT and RoBERTa based on a balance between model capacity and task requirements. Hotel reviews are typically short (1–5 sentences) and written in relatively direct language, in which sentiment signals are conveyed primarily through local lexical and syntactic patterns rather than long-range reasoning. Prior probing studies (e.g., Tenney et al., 2019) suggest that many linguistic features relevant to sentiment classification are captured in the lower and middle layers of transformer models. As DistilBERT preserves much of this representational capacity through knowledge distillation, it remains effective for sentence-level sentiment tasks despite its reduced size (Adel et al., 2022).

In addition, DistilBERT enables faster training and iteration

because of its smaller number of parameters, which is particularly advantageous in a benchmarking setting involving multiple experimental configurations, such as hyperparameter tuning and imbalance handling strategies. Larger models such as BERT and RoBERTa may offer marginal performance gains but also raise the risk of overfitting when applied to moderately sized datasets. Empirically, DistilBERT achieves competitive performance on the test set, supporting Occam's Razor: when a simpler model performs comparably, increased complexity is not justified. The technical details of the distillation training procedure, including the composite loss function and temperature-scaled softmax formulations, are provided in Appendix A.

3.2.2. PhoBert

Unlike English, Vietnamese does not use spaces to define word boundaries and contains tonal markers and compound structures that pose significant challenges for general-purpose tokenizers. As a result, standard BERT models, including multilingual variants, cannot reliably distinguish between syllables and words in Vietnamese, leading to degraded performance on downstream tasks (Thin et al., 2024). To address this, a language-specific model is required.

PhoBERT is the first large-scale monolingual language model dedicated to Vietnamese, available in PhoBERT-base and PhoBERT-large versions (Nguyen & Nguyen, 2020). Built on the RoBERTa training framework (Liu et al., 2019), it incorporates word-level segmentation prior to training to accurately resolve Vietnamese word boundaries, enabling the model to capture morphological, syntactic, and semantic information that syllable-based tokenizers cannot access. Pre-trained on approximately 20 GB of Vietnamese text, PhoBERT has achieved state-of-the-art performance across multiple Vietnamese NLP benchmarks, including part-of-speech tagging, named entity recognition, dependency parsing, and natural language inference, consistently outperforming multilingual alternatives (Nguyen & Nguyen, 2020). These properties make PhoBERT the most appropriate choice for Vietnamese sentiment classification in the present study.

3.2.3. Model deployment

3.2.3.1. Imbalanced data handling. Despite careful preprocessing, model training was hampered by severe class imbalance, with an excess of Positive samples leading to bias toward the Positive label and poor performance on Neutral and Negative classes.

After labeling the datasets as Positive, Neutral, and Negative, we observed a severe imbalance, with Positive samples dominating the distribution. This affected the model training process, as the model tended to predict with a bias toward Positive labels and achieved poor performance on Neutral and Negative labels. Specifically, the label distribution in the English datasets was as follows: Agoda (Positive: 82%, Neutral: 13%, Negative: 5%), Booking (Positive: 92%, Neutral: 5%, Negative: 3%), and TripAdvisor (Positive: 93%, Neutral: 4%, Negative: 3%). For the two Vietnamese datasets, Agoda achieved 80%–13%–7% and Booking achieved 86%–11%–3% for Positive, Neutral, and Negative, respectively.

To mitigate data imbalance, we applied a combined approach of RandomOverSampler and K-Fold Cross-Validation. Instead of discarding excess Positive samples, which would reduce training data and weaken generalization, the method oversampled minority classes (Neutral and Negative) by generating random copies, ensuring balanced label distribution. This allowed the model to learn from a richer set of examples while maintaining equal class representation. RandomOverSampler was chosen over focal loss and class-weighted loss based on its compatibility with transformer fine-tuning dynamics. Because DistilBERT relies on pre-trained representations, modifying the loss function may introduce additional sensitivity to hyperparameters and affect training stability. In particular, focal loss requires tuning of the focusing parameter γ, which adds experimental complexity without clear evidence of consistent gains in moderately imbalanced multi-class settings. Similarly, class-weighted loss can lead to unstable gradient scaling, especially when class distributions are skewed. In contrast, RandomOverSampler addresses class imbalance at the data level without altering the optimization objective. This allows the model to be fine-tuned under more stable and controlled conditions while maintaining reproducibility across experiments. Therefore, it offers a practical and reliable approach for handling imbalance in transformer-based sentiment classification.

In addition, the use of K-Fold Cross-Validation further strengthened the strategy by rotating train-test splits across k folds, ensuring all samples contributed to evaluation, minimizing selection bias, and providing both averaged accuracy and variance metrics. Together, these techniques enhanced classification accuracy across all sentiment labels, reduced bias, and improved the model's stability and reliability in real-world applications.

3.2.3.2. Model deployment and fine-tuning. Within the scope of this study, we implemented and trained the DistilBERT and PhoBERT models, variants of BERT, to perform sentiment analysis on reviews in both English and Vietnamese, as illustrated in Table 3. We used k-fold cross-validation, in which the dataset was divided into k parts, with each fold serving once as validation, while the remaining folds were used for training. For every fold, the model was trained on its respective training set, evaluated on the validation set, and both the model and tokenizer were saved in dedicated directories. The evaluation results, including accuracy and F1 score, were recorded to track progress.

Once training of the sentiment analysis model is complete, we predict on the test set. First, initialize a LabelEncoder (fitted on training labels) to map sentiment labels to integers (Negative→0, Neutral→1, Positive→2) and to inverse-transform predictions back to text. Tokenize the entire test set using the same tokenizer and create a DataLoader with small batches to reduce memory load. To aggregate outputs from k-fold models, initialize probs_sum as a zero array of shape (N_test, num_classes). For each model: load the model, iterate over batches, compute logits → softmax → probabilities for each sample, and add these probabilities into probs_sum at the corresponding sample indices; unload the model if necessary. After processing all k models, compute probs_avg = probs_sum/k. The predicted class for each sample is the argmax of probs_avg, and you can transform indices back to text labels with LabelEncoder.inverse_transform. Also, compute score_difference = top_prob − second_prob on probs_avg to measure confidence. Finally, evaluate predictions against ground-truth labels.

After implementing the model using the proposed research methodology, we conducted experiments on the collected dataset. Throughout training and optimization, we fine-tuned several key hyperparameters to achieve optimal performance.

The training and execution processes of both DistilBERT and PhoBERT models are relatively similar; however, some parameters were fine-tuned specifically for each model. These differences are summarized in the table, clarifying the technical distinctions between the English pre-trained model (DistilBERT) and the Vietnamese pre-trained model (PhoBERT). The tokenizer models, Pre-trained Model Class, and Pre-trained Model Checkpoint were selected to match the target language that each model processes. In addition, the two most distinctive fine-tuning parameters are the Max length and n_splits of the K-Fold Cross Validation technique.

Table 3
Comparison of Training Configurations: DistilBERT vs. PhoBERT.

Process		DistilBERT	PhoBERT
Data Tokenization	Tokenizer model	DistilBertTokenizer	PhoBertTokenizer
	Max length	512	256
Model Training	Pretrained Transformer Models	DistilBertForSequenceClassification	AutoModelForSequenceClassification
	Pretrained model names	distilbert-base-uncased	vinai/phobert-base
	K-Fold Cross Validation	n_splits = 5	n_splits = 2
	Epochs	num_train_epochs = 3	
	Learning rate	learning_rate = 2e-5	
	Batch size	batch_size = 16	
	Logging	logging_steps = 10	
	Weight decay	weight_decay = 0.01	

Specifically, with DistilBERT, the Max length was set to 512 tokens, while PhoBERT only used 256 tokens. This reflects the concise and information-dense nature of Vietnamese. The shorter length conserves resources and enhances focus on key contextual nuances.

DistilBERT for English applies 5-fold cross-validation, while PhoBERT for Vietnamese uses 2-fold. This difference arises from linguistic complexity and dataset scale. English, with diverse grammar and a large dataset (>100,000 samples), benefits from 5-fold cross-validation to enhance generalization. In contrast, Vietnamese has simpler structures and a smaller dataset (~10,000 samples), so 2 folds are sufficient to maintain balance between training and validation while ensuring efficiency.

In summary, the number of folds is not a fixed value for all cases but must be flexibly adjusted based on data characteristics and computational capabilities. Selecting the appropriate number of folds is the result of research, experimentation, and model optimization with different values. By using this number of folds, the team observed that the model effectively exploits language features while ensuring efficient operation without consuming excessive resources.

3.3. Topic modeling

3.3.1. BERTopic model

BERTopic is a topic modeling technique based on BERT that represents text as semantic embeddings rather than discrete words (Grootendorst, 2022). Specifically, it uses Sentence-BERT to generate document embeddings, which are then reduced in dimensionality using UMAP and clustered with HDBSCAN, a density-based algorithm capable of detecting clusters of varying density. To extract representative keywords for each cluster, BERTopic uses class-based TF-IDF (c-TF-IDF) instead of traditional TF-IDF. The weight of a word t in a cluster c is calculated as:

$$W_{t,c} = tf_{t,c} \cdot log\left(1 + \frac{A}{tf_t}\right) \quad (6)$$

where $tf_{t,c}$ denotes the frequency of word ttt in documents belonging to the cluster c, A is the average number of words per topic, and tf_t is the total frequency of t across the entire dataset. This approach allows BERTopic to identify more distinctive keywords for each topic and reduce redundancy across clusters. Additionally, after training a BERTopic model, users can reduce the number of topics, thereby enabling the refinement of specific topics based on the actual number generated (Sánchez-Franco & Rey-Moreno, 2022).

The model's performance depends on several hyperparameters: the number of neighbors and components in UMAP influence the compactness of clusters; the minimum distance defines separation between topics; the minimum topic size controls granularity; the number of top words determines interpretability; and parameters such as n-gram range, min_df, and max_df filter out noise while enhancing topic quality. Proper tuning of these hyperparameters ensures a balance between topic coherence, specificity, and conciseness.

In this study, the BERTopic model was used to uncover hidden topics in two datasets corresponding to the Vietnamese and English languages. Because of the distinct differences in data volume and linguistic characteristics between the two languages employed in this study, the BERTopic model was trained separately for each language. For each training session, the model's parameters and hyperparameters were carefully adjusted to optimize the results for the specific language. For English texts, the all-mpnet-base-v2 embedding model was used to ensure accurate semantic representation, as shown in Table 4. For Vietnamese texts, the study used paraphrase-multilingual-MiniLM-L12-v2, a lightweight multilingual model optimized for efficiency and suitable for resource-limited environments (see Table 5).

To evaluate the quality of the extracted topics, both quantitative coherence metrics and qualitative validation were used. From a quantitative perspective, topic coherence scores, including CV Coherence and NPMI Coherence, are used to measure the semantic consistency of keywords within each topic. The CV coherence score is a metric based on the co-occurrence of words within a given context. It uses cosine similarity between word distributions, combined with semantic space models (e.g., word embeddings), to assess the degree of connectivity between words within a topic. In contrast, the Normalized Pointwise Mutual Information (NPMI) coherence score captures conditional co-occurrence statistics of words, making it particularly suitable for evaluating word associations in large corpora. It is an extension of Pointwise Mutual Information (PMI), normalized to the range [−1; 1] to enhance comparability across different topic models. Higher NPMI values indicate that words within a topic tend to co-occur more often than would be expected by chance, reflecting stronger semantic association. By combining both CV and NPMI, the evaluation balances semantic embedding similarity with reliable statistical co-occurrence, thus ensuring a more comprehensive assessment of topic quality.

Coherence scores provide only a numerical measure and may not fully capture the contextual appropriateness of the topics. Therefore, the evaluation was further supported by qualitative validation through cross-checking and topic verification. In this process, important keywords were first identified using c-TF-IDF, where the keywords with the highest scores were considered representative of their respective topics. These representative keywords were then validated through manual examination of Representative Documents, ensuring that the extracted topics were not only technically sound but also meaningful in their specific context. This combination of quantitative and qualitative evaluation provides a comprehensive assessment, ensuring that

Table 4
BERTopic Configuration and Parameters for English text.

Process		BERTopic Parameters
Dimensionality Reduction (UMAP)	Number of neighbors	n_neighbors = 15
	Target space dimensions	n_components = 5
	Minimum distance between points	min_dist = 0.1
	Distance metric	metric = 'cosine'
	Random state	random_state = 100
Embedding Model	Sentence Transformer model	embedding_model = SentenceTransformer("all-mpnet-base-v2")
TF-IDF Vectorization	N-grams	ngram_range=(1,2)
	Minimum document frequency	min_df = 0.02
	Maximum document frequency	max_df = 0.95
BERTopic Parameters	Minimum topic size	min_topic_size = 120
	Number of top words	top_n_words = 10
	Language	language = "english"
	Low memory mode	low_memory = True
	Calculate topic probabilities	calculate_probabilities = True

Table 5
BERTopic Configuration and Parameters for Vietnamese text.

Process		BERTopic Parameters
Dimensionality Reduction (UMAP)	Number of neighbors	n_neighbors = 10
	Target space dimensions	n_components = 5
	Minimum distance between points	min_dist = 0.1
	Distance metric	metric = 'cosine'
	Random state	random_state = 100
Embedding Model	Sentence Transformer model	embedding_model = SentenceTransformer("paraphrase-multilingual-MiniLM-L12-v2")
TF-IDF Vectorization	N-grams	ngram_range=(1,2)
BERTopic Parameters	Minimum topic size	min_topic_size = 50
	Number of top words	top_n_words = 10
	Language	language = "multilingual"
	Low memory mode	low_memory = True
	Calculate topic probabilities	calculate_probabilities = True

the topics are both statistically valid and practically interpretable.

3.3.2. Comparison with LDA

Latent Dirichlet Allocation (LDA) is a probabilistic generative model that assumes each document is a mixture of topics, and each topic is a distribution over words. Its primary goal in this research is to uncover interpretable topics using a traditional statistical approach before assessing the effectiveness of embedding-based models such as BERTopic. The input data for LDA consisted of preprocessed customer reviews, tokenized into word lists, and represented using the Bag-of-Words (BoW) method. A vocabulary dictionary was constructed, and each document was encoded as a frequency vector of words. This format allowed the model to focus on word co-occurrence patterns, abstracting away from word order or context.

The experiment was conducted on a hotel review dataset comprising reviews from three platforms. Because the data from each platform individually was not large enough to train the model effectively, the datasets were aggregated to form a sufficiently sized combined dataset, enabling the LDA model to learn topics more distinctly. After preprocessing, texts shorter than 5 words were removed to ensure the quality of the model input. A Gensim Dictionary was created, in which words appearing in fewer than 5 documents or more than 70% of the total documents were removed to reduce noise. The model was then trained with different numbers of topics (ranging from 3 to 15) to determine the optimal topic count. Training parameters included passes = 20 and iterations = 100 to ensure stable convergence, alpha = 'auto' and eta = 'auto' for adaptive topic-word distributions, and minimum_probability = 0.001 to filter out low-probability terms within topics.

After training, the model yielded a set of main topics characterized by top keywords and their probabilities. The results were evaluated using two key metrics: Perplexity, measuring the model's ability to generalize to unseen data (lower is better), and Coherence Score (C_V), assessing the semantic similarity of the top words within each topic (higher is better).

To provide a consistent point of reference, both LDA and BERTopic were evaluated using common coherence metrics, particularly C_V and C_NPMI, which are widely used in topic modeling research. While LDA relies on probabilistic modeling over Bag-of-Words representations, BERTopic uses Transformer-based embeddings combined with density-based clustering in semantic space. As a result, the comparison is intended to serve as a baseline-oriented assessment rather than a direct equivalence between model architectures. Given these differences, the analysis incorporates both quantitative evaluation (coherence scores) and qualitative inspection (manual validation of topic relevance) to provide a more comprehensive understanding of topic quality. This dual perspective allows the study to capture not only statistical patterns but also the practical interpretability and usefulness of the extracted topics for real-world applications.

4. Results and evaluation

4.1. Result in sentiment analysis

The results in Table 6 show that the proposed Transformer-based models, DistilBERT and PhoBERT, significantly outperform traditional Naive Bayes and SVM baselines across all evaluation metrics and datasets. DistilBERT achieved exceptional accuracy, peaking at 0.9940 on the TripAdvisor dataset, while PhoBERT reached a strong 0.8867 on Vietnamese reviews, both showing substantial improvements over the baseline models. Furthermore,

Table 6
Comprehensive performance comparison on validation datasets.

Dataset	Model	Accuracy	Precision	Recall	F1-Score
English Reviews					
Agoda (EN)	Naive Bayes	0.8240	0.8150	0.8240	0.8190
	SVM	0.8670	0.8600	0.8670	0.8630
	DistilBERT	**0.9571**	**0.9593**	**0.9571**	**0.9570**
Booking (EN)	Naive Bayes	0.8450	0.8380	0.8450	0.8410
	SVM	0.8820	0.8750	0.8820	0.8780
	DistilBERT	**0.9879**	**0.9892**	**0.9879**	**0.9879**
TripAdvisor (EN)	Naive Bayes	0.8510	0.8420	0.8510	0.8460
	SVM	0.8900	0.8840	0.8900	0.8870
	DistilBERT	**0.9940**	**0.9941**	**0.9940**	**0.9940**
Merged (EN)	Naive Bayes	0.8300	0.8250	0.8300	0.8270
	SVM	0.8750	0.8680	0.8750	0.8710
	DistilBERT	**0.9850**	**0.9854**	**0.9850**	**0.9848**
Vietnamese Reviews					
Agoda (VI)	Naive Bayes	0.7650	0.7500	0.7650	0.7570
	SVM	0.8020	0.7950	0.8020	0.7980
	PhoBERT	**0.8867**	**0.8863**	**0.8867**	**0.8862**
Agoda + Booking (VI)	Naive Bayes	0.7580	0.7420	0.7580	0.7500
	SVM	0.7950	0.7880	0.7950	0.7910
	PhoBERT	**0.8849**	**0.8849**	**0.8849**	**0.8845**

the negligible difference between precision and recall (e.g., 0.9941 vs. 0.9940 for TripAdvisor) indicates high model stability and effective handling of sentiment class distribution, confirming the superiority of pre-trained language models in capturing complex semantic nuances in hospitality feedback.

4.1.1. English reviews using the DistilBERT model

For the English dataset, the DistilBERT model was evaluated on four corpora: Agoda, Booking, TripAdvisor, and a merged dataset.

As shown in Fig. 3, validation results indicate uniformly strong performance, with accuracy, precision, recall, and F1 exceeding 0.95 across cases. TripAdvisor attained the highest aggregate performance (metrics > 0.99); Booking and the merged corpus followed closely at 0.98; and Agoda yielded the lowest validation scores at approximately 0.95. Recall ranged from 0.9571 to 0.9940, suggesting strong sensitivity to true labels and implying greater consistency in TripAdvisor data and greater linguistic variability in Agoda.

In Fig. 4, label-wise analysis reveals that DistilBERT reliably detects positive sentiment, with accuracies above 0.90 across all datasets, peaking at 0.98 on TripAdvisor and Booking. In contrast, negative and neutral classes exhibit substantially lower discriminative performance: negative accuracy spans approximately 0.47–0.73 (best on TripAdvisor), while neutral accuracy is weakest from 0.27 to 0.60, lowest on Booking. Training on the merged corpus improves generalization for positive labels (0.96) and yields moderate gains for negative (0.62) and neutral (0.49).

As shown in Fig. 5, DistilBERT consistently achieves high accuracy for positive reviews (from 91% to 98%), while negative and neutral labels remain difficult to distinguish. On Agoda, neutral detection is relatively stronger (about 60%), though negative reviews are often misclassified as neutral. Booking shows the weakest performance, with negative accuracy around 47%, and neutral is frequently confused with positive (around 69%). TripAdvisor attains near-perfect recognition of positive sentiment (≈98%) but still exhibits notable confusion between negative and neutral categories. On the merged dataset, positive accuracy remains high (at 96%), yet neutral reviews are correctly identified in fewer than half of cases, with substantial misclassification as positive. Overall, the model reliably detects positive sentiment but struggles with ambiguous expressions in negative and neutral reviews, highlighting the need for further refinement.

4.1.2. Vietnamese reviews using the PhoBERT model

For Vietnamese data, the PhoBERT model was trained and validated on two datasets: Agoda and a combined Agoda–Booking corpus.

As shown in Fig. 6, overall performance was stable across both settings, with accuracy, precision, recall, and F1-scores consistently in the range of 0.88 to 0.89. On Agoda, accuracy reached 0.8867, while the combined dataset yielded 0.8849, suggesting that adding Booking data introduced only marginal variation. Recall also remained stable at approximately 0.88 across both datasets, indicating reliable coverage of true sentiment labels despite stylistic differences in user expression.

Label-level analysis, illustrated in Fig. 7, shows that positive sentiment was recognized with the highest accuracy (85% for Agoda and 83% for Agoda–Booking), reflecting the clearer linguistic markers of satisfaction in Vietnamese reviews. By contrast, performance was weaker for negative sentiment (around 58%) and moderate for neutral (from 64% to 66%), with frequent misclassification between these two categories. These results show

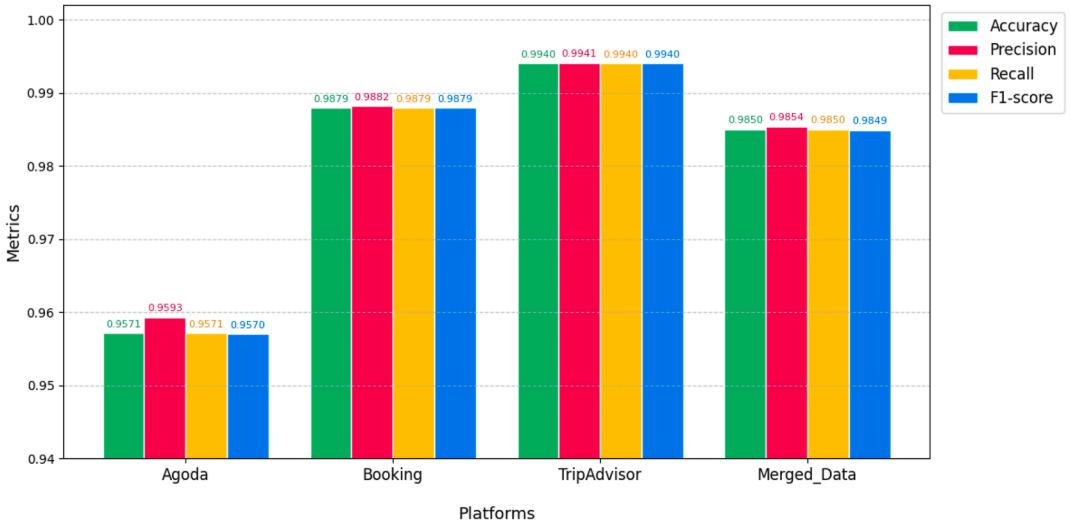


Fig. 3. Performance of the DistilBERT model on the validation dataset.

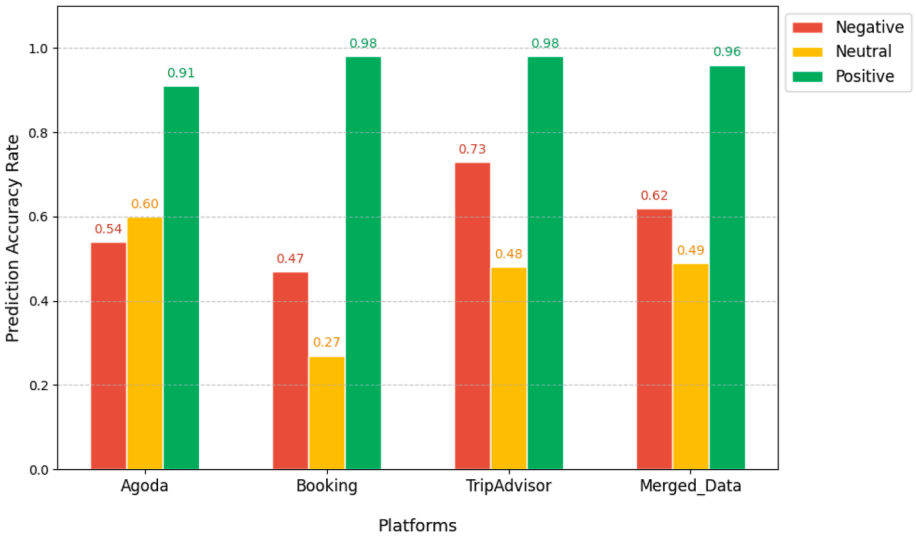


Fig. 4. Accurate classification rates of labels by dataset.

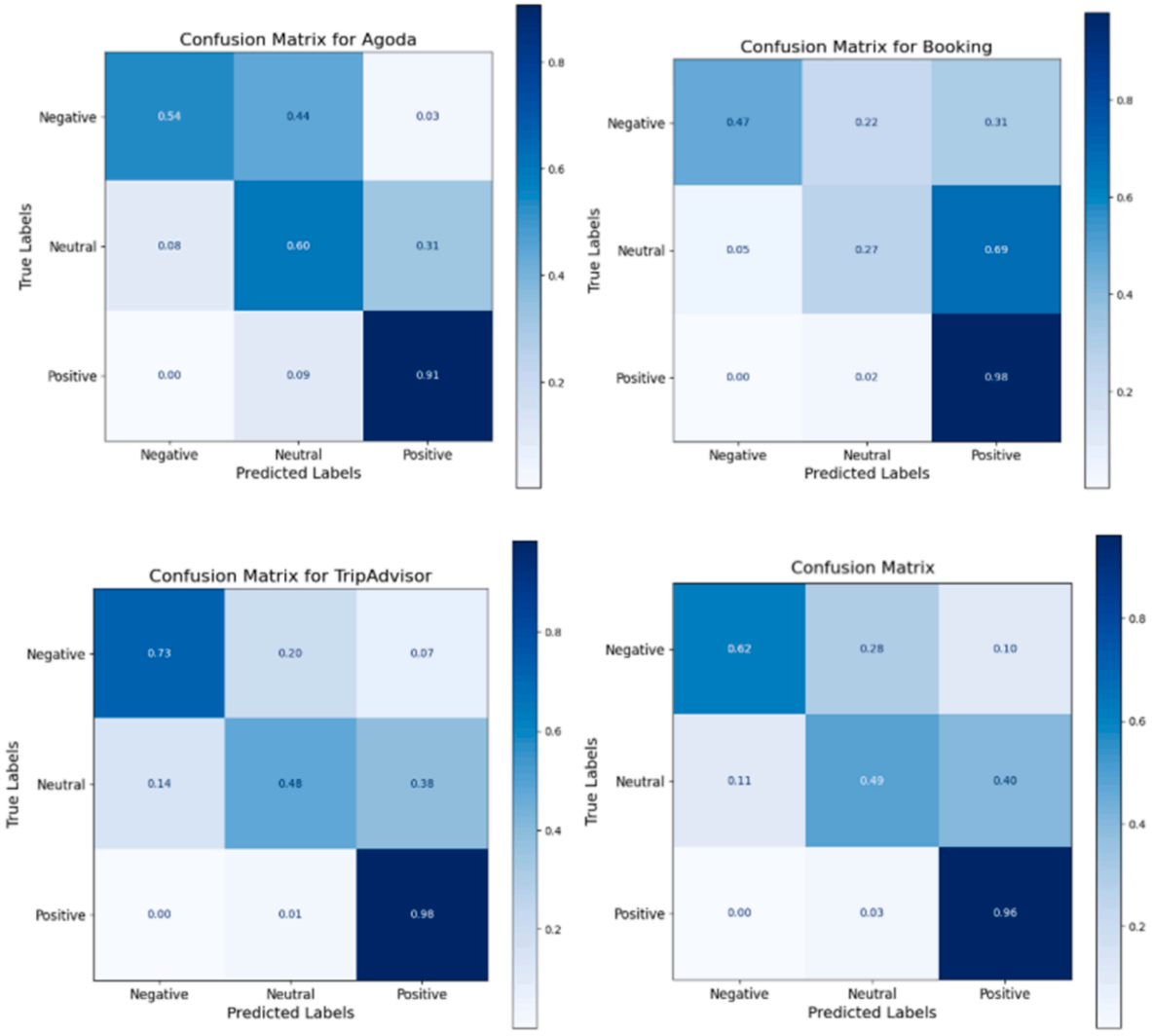


Fig. 5. Confusion matrices for datasets - DistilBERT model.

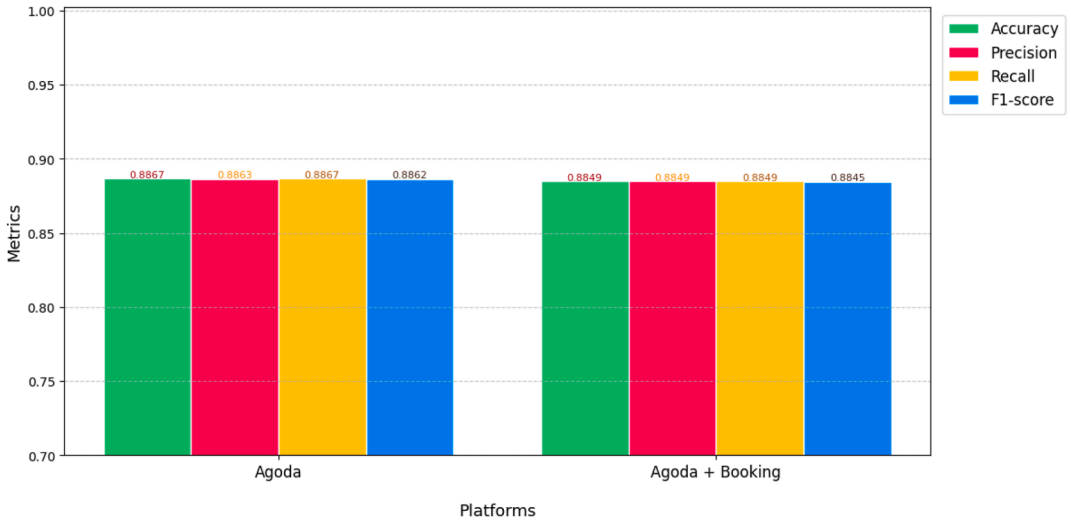


Fig. 6. Performance of PhoBERT model on the validation dataset.

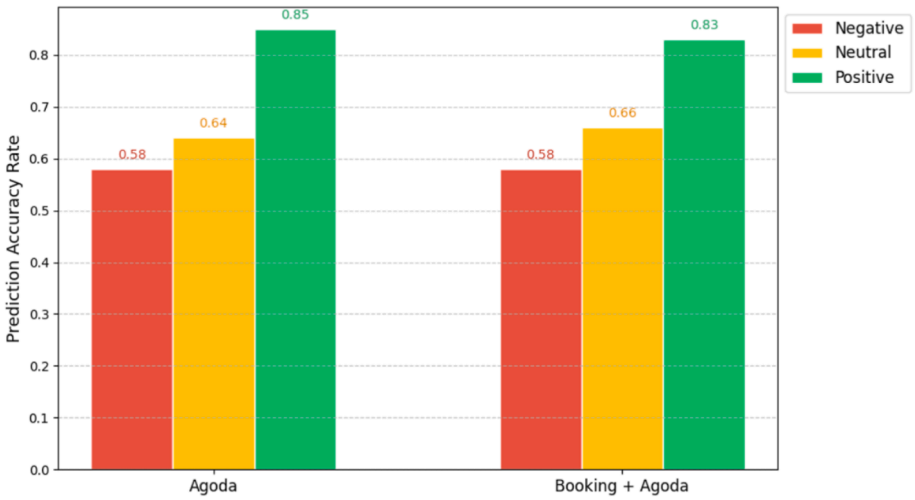


Fig. 7. The accurate classification rate of labels by dataset.

PhoBERT's strength in capturing explicit positive expressions while highlighting its limitations in distinguishing ambiguous or indirectly expressed negative and neutral sentiments.

As shown in Fig. 8, PhoBERT shows strong performance in classifying positive sentiment, with accuracy consistently around 83% to 85%, reflecting clear lexical cues of satisfaction in Vietnamese reviews. In contrast, negative sentiment is correctly identified in only about 58% of cases, with frequent misclassification as neutral. Neutral sentiment shows moderate recognition (around 64% to 66%) but is often confused with positive feedback, highlighting ambiguity in emotionally subtle exprssions. Overall, the model reliably detects positive sentiment but struggles with negative and neutral categories, highlighting the need for further refinement to address linguistic ambiguity.

4.2. Result in topic modeling

4.2.1. BERTopic

For the Vietnamese dataset, BERTopic initially extracted 41 topics, later refined to 18 after applying the reduce_topics function and manual inspection (see Fig. 9). These topics capture diverse dimensions of customer experience. Several highlight positive impressions, such as Topic 0 (High Satisfaction) with repeated mentions of “mọi thứ đều tốt” and “tuyệt vời từ phòng đến nhân viên,” Topic 1 (Beach & Pool) emphasizing beach access and scenic views, Topic 2 (Breakfast & Food) describing buffet variety and food quality, Topic 5 (Rooms & Cleanliness) praising aesthetics and comfort, and Topic 15 (Holiday & Thanks) capturing memorable vacations. In contrast, service-related feedback is reflected in Topic 3 (Staff & Rooms) and Topic 4 (Complaints), which mix comments on staff attitude and room service, as well as Topic 8 (Check-In) and Topic 16 (Slow Service), highlighting delays, crowded elevators, and dissatisfaction with response times.

Other topics point to specific facilities and amenities. Topic 6 (Bathroom & Odor) highlights issues with water pressure, hot water, and unpleasant smells, while Topic 9 (Parking) addresses availability and fees. Topic 10 (Noise & Sleep) reveals disturbances from construction and neighbors, Topic 12 (TV & Entertainment) reflects expectations for modern in-room streaming services, Topic 13 (Wi-Fi & Connection) shows persistent complaints about unstable networks, and Topic 14 (View & Hygiene) captures mixed opinions about promised versus actual views. Finally, broader

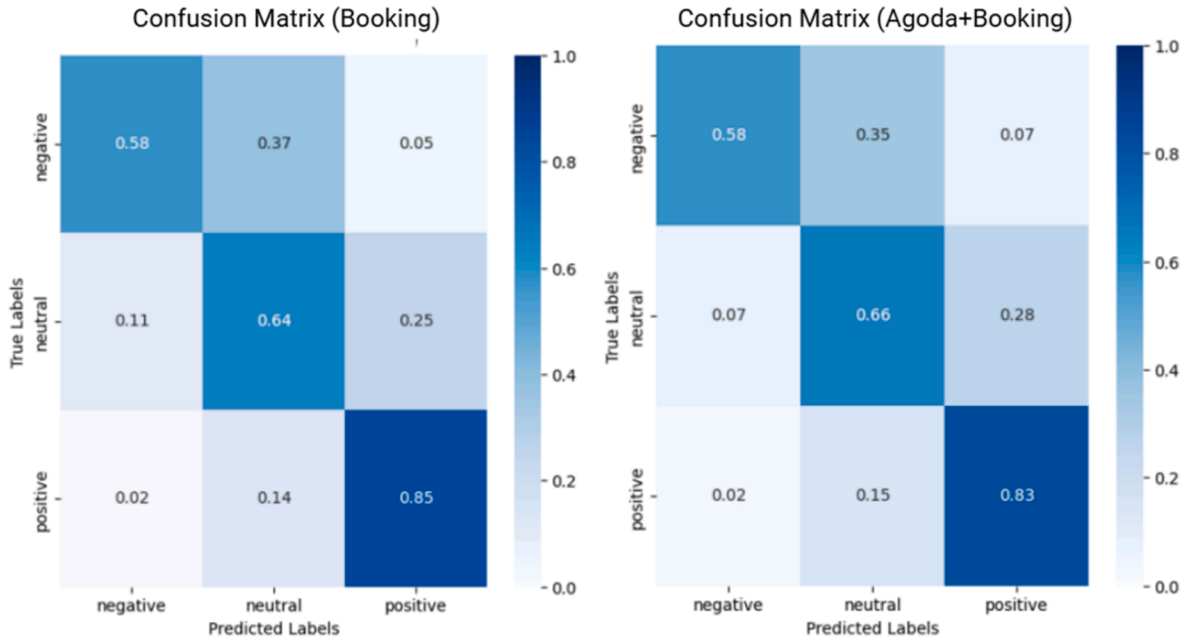


Fig. 8. Confusion matrix for datasets - PhoBERT model.

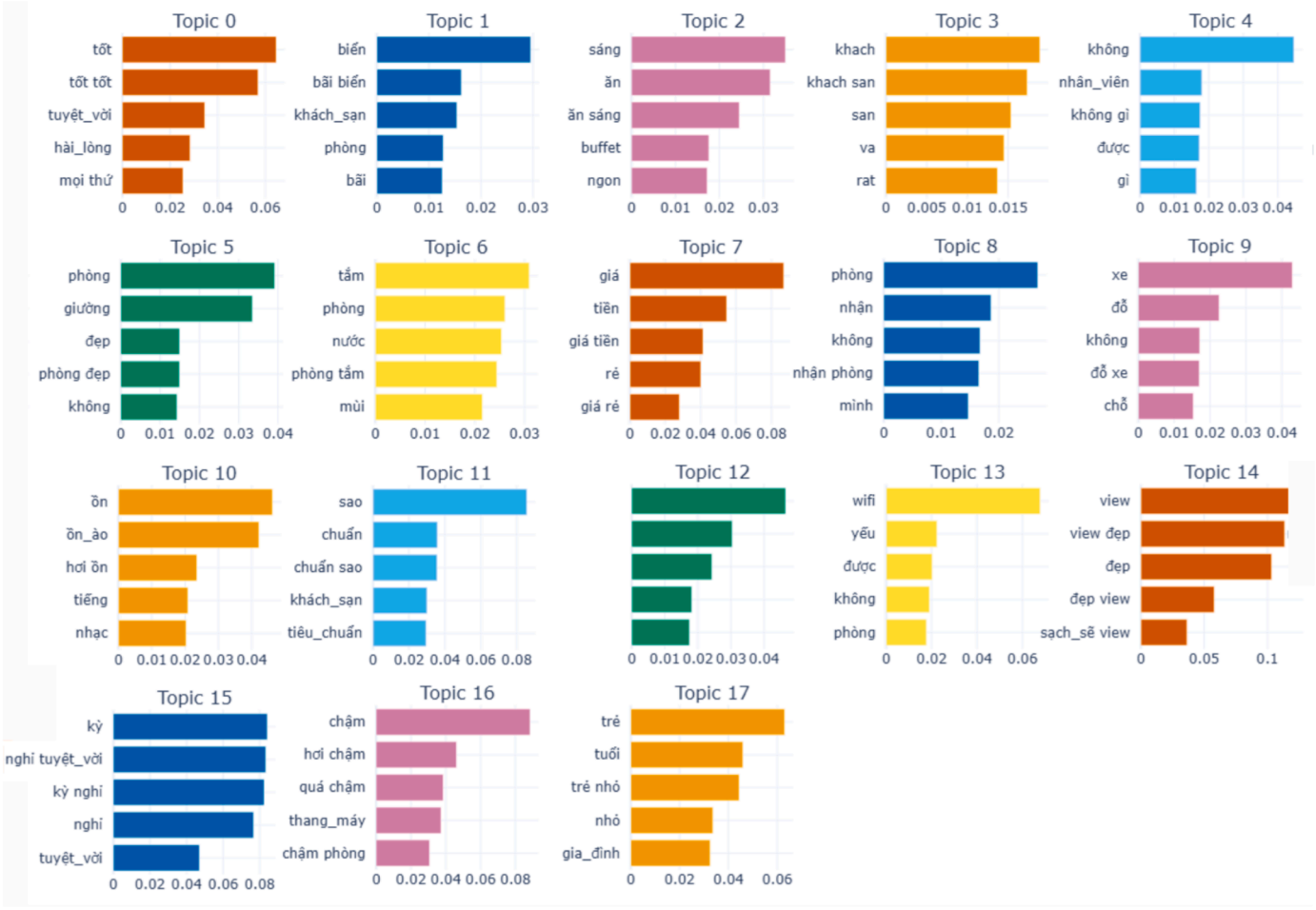


Fig. 9. Top keywords with the highest c-TF-IDF scores in each topic (Vietnamese).
Source: Authors

concerns are evident in Topic 7 (Price & Value), contrasting perceptions of affordability and overpriced services; Topic 11 (Hotel Rating), questioning whether hotels meet their star standards; and Topic 17 (Kids & Families), noting the importance of safe, family-friendly amenities. Together, the 18 topics provide a comprehensive overview of both positive highlights and critical pain points in hotel services.

For English data, the model first identified 61 topics. After refinement, 41 remained, which were further grouped into 8 broader content categories based on keyword similarity and semantic overlap.

As shown in Fig. 10, the first set of categories covers service quality, accessibility, and physical comfort. Group Service and Staff (Topics 0, 1, 2, 5, 15, 25) highlights interactions with employees, with keywords such as friendly, helpful, professional, and cleanliness frequently associated with positive feedback, while Topic 5 reveals dissatisfaction with staff responsiveness. Group Location and Accessibility (Topics 1, 7, 16, 20, 35) highlights proximity to landmarks such as the Old Quarter, Ben Thanh Market, and the airport, with Topic 35 emphasizing walkability to tourist attractions as a major satisfaction factor. Group Room Amenities and Facilities (Topics 6, 9, 10, 19, 22, 27, 29, 30) addresses practical elements including bathrooms, noise, bed comfort, Wi-Fi, air conditioning, elevators, gym, and cleanliness, with repeated mentions of noisy rooms and weak Wi-Fi indicating recurrent pain points.

The remaining categories capture experiential, emotional, and evaluative dimensions. Group Overall Experience and Emotions (Topics 17, 23, 37, 40) conveys broad impressions using expressions such as highly recommended, nothing, everything, reflecting polarized views of satisfaction or disappointment. Group Food and Dining (Topics 4, 11, 12, 32) highlights breakfast quality, spa experiences, rooftop dining, and cultural cuisines (e.g., Japanese food), which strongly shape customer impressions of their stay. Luxury Hotels and Resorts (Topics 3, 13, 28, 36, 38, 39) centers on premium brands such as Vinpearl and Silverland, emphasizing beach access, scenic views, and private services. Group Price and Value (Topics 21, 24) reflects perceptions of value for money, where terms like great value contrast with complaints about overpriced services. Finally, Group Accommodation Types and Special Occasions (Topics 14, 18, 31, 33, 34) highlights varied lodging types such as serviced apartments or hostels, along with occasions such as honeymoons and birthdays, pointing to the role of personalization in guest satisfaction.

The 41 topics extracted from English reviews present a comprehensive picture of customer experience, spanning operational factors (service quality, location, amenities) and subjective impressions (overall emotions, special occasions). Positive topics frequently mention helpful staff, good breakfast, comfort, and luxury ambience, whereas negative clusters focus on noise, weak Wi-Fi, slow elevators, and mismatched value for price. This balance between satisfaction and complaints confirms BERTopic's capacity to capture both granular service details and overall perceptions, offering actionable insights for hospitality providers to enhance both quality and guest experience.

From a quantitative perspective, BERTopic achieved stable performance across both datasets, with average coherence scores of 0.5597 (c_v) and 0.0579 (c_npmi) for Vietnamese, and 0.5524 (c_v) and 0.0498 (c_npmi) for English. These similar values highlight the model's stability in handling multilingual feedback. At the topic level, coherence scores varied: in the Vietnamese dataset, high-performing topics such as Topic 3 (c_v = 0.72, c_npmi = 0.21), Topic 8 (c_v = 0.67), and Topic 15 (c_v = 0.69, c_npmi = 0.23) reflected strong semantic focus, particularly on aspects such as staff service or check-in procedures. By contrast, topics concerning noise, pricing, or infrastructure showed moderate or lower coherence, sometimes with slightly negative c_npmi, indicating broader scopes or fragmented customer expressions. Overall, these results confirm the model's ability to detect coherent themes while revealing variability across specific service-related dimensions.

4.2.2. Comparison

The comparison is based on four key criteria: topic coherence, interpretability, stability, and computational efficiency.

BERTopic achieved a higher C_V score (0.5524) than LDA (0.5218), indicating that BERTopic produced more coherent and semantically consistent topics. Although its C_NPMI score (0.0498) was slightly lower than that of LDA (0.0697), this difference should be interpreted with caution, as C_NPMI is fundamentally grounded in word co-occurrence statistics and may not fully capture the contextual semantic relationships modeled by embedding-based approaches. As such, the observed gap does not necessarily reflect inferior topic quality, but rather highlights a divergence between evaluation metrics and modeling paradigms. A lower C_NPMI score does not necessarily indicate inferior topic quality, but rather reflects a difference in how topic coherence is measured. Embedding-based models may produce semantically richer and more interpretable topics that are not fully aligned with co-occurrence-based evaluation metrics.

In terms of interpretability, BERTopic showed a clear advantage. By using embeddings and clustering, it generated meaningful phrase-based topics such as “friendly staff” and “customer support,” which enabled faster and more intuitive topic labeling. In contrast, LDA predominantly produced individual keywords (e.g., “hotel,” “service,” “staff”), making it more challenging to infer the overall theme. For instance, in topics related to hotel service, BERTopic yielded semantically complete expressions such as “excellent service” and “professional team,” whereas LDA provided fragmented lexical signals that are less directly interpretable.

BERTopic also showed superior stability when the model was rerun on the same dataset, producing consistent clusters due to its reliance on embeddings and density-based clustering. LDA, on the other hand, showed greater variability across runs because of its dependence on Gibbs Sampling. Despite these advantages, LDA retained a comparative strength in computational efficiency, offering faster training times and lower resource requirements, which may be preferable for very large-scale datasets or environments with constrained computational capacity.

In addition, to provide a more rigorous assessment against state-of-the-art architectures, Contextualized Topic Models (CTM) were incorporated into the comparison. As shown in Table 7, while the Contextualized Topic Model (CTM) presents a slight improvement in coherence scores over LDA and matches some of BERTopic's metrics, BERTopic is selected as the optimal architecture for this research. The primary advantage lies in its modular approach, which avoids the limitations of Bag-of-Words representations inherent in both LDA and CTM. BERTopic effectively captures the linguistic nuances of bilingual hotel reviews, producing topics that are not only statistically coherent but also highly interpretable for practitioners. For instance, whereas CTM occasionally generated overlapping terms across service dimensions, BERTopic maintained clear boundaries between themes such as 'amenities', 'location', and 'staff responsiveness', which is vital for actionable insights in hospitality management.

5. Conclusion

This study makes substantial contributions to sentiment analysis and topic modeling within the hospitality domain by



Fig. 10. Top keywords with the highest c-TF-IDF scores in each topic.
Source: Authors

Table 7
Comparison table of coherence scores between BERTopic, LDA and CTM.

Model	C_V	C_NPMI
LDA	0.5218	0.0697
BERTopic	0.5524	0.0498
CTM	0.5341	0.0615

integrating transformer-based architectures and large-scale multilingual review data. Beyond model integration, the study contributes to both theoretical understanding and methodological design in multilingual NLP for tourism analytics. In contrast to earlier research that predominantly used classical machine-learning or lexicon-based approaches, such as Naive Bayes, SVM,

and LSTM models, which often fail to capture subtle emotional nuances (Le & Huh, 2021), the present work shows that transformer models, DistilBERT for English and PhoBERT for Vietnamese, offer superior capability in representing contextual meaning. Nonetheless, the results also reveal persistent challenges in distinguishing neutral from negative sentiment, suggesting that even state-of-the-art models may require additional fine-tuning or task-specific adaptations for sentiment-rich domains.

From a theoretical perspective, the findings provide empirical evidence of cross-linguistic differences in sentiment expression, indicating that English and Vietnamese users differ in how dissatisfaction is expressed in terms of linguistic intensity and structure. This highlights the importance of culturally and linguistically aware sentiment analysis in multilingual contexts.

A key methodological contribution of this research lies in its handling of the substantial class imbalance inherent in hotel review datasets. While prior studies frequently rely on SMOTE for imbalance correction (Singgalen, 2024), this study shows that a combination of RandomOverSampler and K-fold cross-validation yields more stable and reliable performance when applied to transformer models. This finding highlights the importance of selecting imbalance-handling techniques that align with the underlying modeling architecture, particularly for text-based deep learning tasks. Rather than treating imbalance purely as a preprocessing step, the study highlights it as a practical challenge in real-world sentiment analysis that requires careful handling when using deep learning models.

In the domain of topic extraction, the study addresses well-documented limitations of Latent Dirichlet Allocation, including its reliance on fixed topic numbers and its inability to incorporate contextual semantics (Abuzayed & Al-Khalifa, 2021). By employing BERTopic, an embedding- and clustering-based approach, the analysis captures more coherent, flexible, and interpretable themes across both English and Vietnamese datasets. The results suggest that embedding-based topic models are more effective at capturing fine-grained service attributes in multilingual settings compared to traditional probabilistic approaches.

Despite these contributions, several limitations should be acknowledged. The Vietnamese dataset remains relatively small, which may limit model generalizability across different linguistic groups. The strong positivity bias observed across all platforms also complicates sentiment prediction, particularly for minority classes. Moreover, although sentiment analysis and topic modeling were conducted effectively, they were implemented as separate processes. Their integration, for example, through aspect-based sentiment analysis, could yield deeper insight into how different service attributes contribute to positive or negative experiences. Topic coherence metrics, while acceptable, also indicate potential for improvement through domain-adapted embeddings or transformer-based topic modeling methods.

Future research should explore semi-supervised and weakly supervised learning strategies to improve labeling quality beyond platform-derived rating heuristics. Integrating aspect-based sentiment analysis (ABSA) may allow more precise mapping between sentiment polarity and specific service dimensions (Carrasco & Dias, 2024; Sann & Lai, 2020; Wu et al., 2019). Incorporating demographic or contextual metadata could further enable traveler segmentation and behavioral insights. Finally, applying this methodological framework to other industries, such as finance, healthcare, or education, may extend its impact and validate its robustness across domains.

Overall, by combining transformer-based sentiment classification with advanced semantic topic modeling, this study offers a comprehensive and scalable framework for understanding customer perceptions in Vietnam's hospitality sector. The study contributes both empirically and methodologically by uncovering multilingual sentiment patterns and providing practical design insights for handling real-world data challenges. The findings advance methodological practice while offering actionable insights to help hotel managers enhance service quality, customer satisfaction, and strategic decision-making.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) for language editing in English-American style. After using this tool, the authors reviewed and revised the text carefully, and take full responsibility for the content of this publication.

Acknowledgements

This research is funded by the University of Economics and Law, Vietnam National University Ho Chi Minh City, Vietnam.

APPENDIX

A. DistilBERT: Knowledge Distillation and Training Objectives

DistilBERT is trained using a knowledge distillation process in which a smaller student model learns to replicate the behavior of a larger teacher model (BERT) through a composite loss function combining three components (Sanh et al., 2019):

$$L = \alpha \,.\, L_{mlm} + \beta \,.\, L_{ce} + \gamma \,.\, L_{cos} \tag{1}$$

where the coefficients α, β, and γ control the relative contribution of each loss component during training.

In this context, L_{mlm} represents the language modeling loss, which is typically defined as the cross-entropy loss between the predicted distribution and the actual distribution of the masked words. This loss helps the model learn how to predict the masked words in a sentence.

$$L_{mlm} = -\sum_{i} p(i)\ log(q(i)) \tag{2}$$

In this context, $p(i)$ and $q(i)$ are the actual probability and predicted probability for the i-th word in the sentence, which represents the actual distribution the model needs to learn and the probability for each word in the vocabulary based on the context surrounding the masked word.

Next, the distillation loss function (L_{ce}) aims to generate a rich training signal by leveraging the entire distribution of the teacher model:

$$L_{ce} = \sum_{i} t_i\ log(s_i) \tag{3}$$

With t_i (corresponding to s_i) being the probability estimated by the teacher model (corresponding to the student model).

To compute the probabilities for the two loss functions L_{mlm} and L_{ce}, a softmax model (Hinton et al., 2015) is used, where the probabilities are calculated by fine-tuning the temperature parameter (T), which adjusts the "softness" or "hardness" of the probability distribution.

$$p_i = \frac{exp(z_i/T)}{\sum_j exp\left(z_j/T\right)} \quad (4)$$

The parameter T allows adjustment of the confidence level of the model in its predictions. When T = 1, the probability distribution is computed entirely based on the raw logits. When T > 1, the distribution becomes "softer," meaning the probabilities between classes are closer to each other, making the model less confident in its predictions. This is useful in tasks like text generation, where diversity in the output is desired. On the other hand, when T < 1, the distribution becomes "harder," with one class having a much higher probability compared to others, making the model more confident in its prediction. Typically, in DistilBERT applications, T is set to 1 during the inference phase to restore the standard softmax.

Additionally, the cosine loss function (L_{cos}) helps align the direction of the hidden state vectors between the student and teacher models, thereby maintaining high performance while reducing the model size.

$$L_{cos} = 1 - \frac{h_{student} \cdot h_{teacher}}{||h_{student}||\,||h_{teacher}||} \quad (5)$$

B. PhoBERT: Tokenization and Pre-training Procedure

PhoBERT addresses the core challenge of Vietnamese word segmentation through a two-stage tokenization process prior to model training (Nguyen & Nguyen, 2020). In the first stage, raw Vietnamese text is segmented using RDRSegmenter from the VnCoreNLP toolkit (Nguyen et al., 2017; Vu et al., 2018), which identifies compound word boundaries by concatenating constituent syllables with an underscore character "_" (e.g., "học_sinh" for "student"). This step is essential because Vietnamese does not use whitespace to delimit words, and naive whitespace tokenization would incorrectly treat each syllable as an independent token, resulting in a loss of morphological and semantic information.

In the second stage, the segmented text is encoded using Byte-Pair Encoding (BPE) at the word level rather than the syllable level as in standard BERT. BPE iteratively merges the most frequently co-occurring symbol pairs in the training corpus, producing a subword vocabulary that balances lexical coverage with token efficiency. By applying BPE at the word level after RDRSegmenter preprocessing, PhoBERT substantially reduces the total number of tokens per document compared to syllable-level tokenization, thereby improving both training efficiency and downstream task performance.

With respect to pre-training objectives, PhoBERT follows the RoBERTa framework (Liu et al., 2019) by removing the Next Sentence Prediction (NSP) task used in original BERT and training exclusively with the Masked Language Model (MLM) objective. This decision is motivated by empirical findings showing that NSP provides limited benefit and can introduce noise into the pre-training signal. PhoBERT also adopts larger batch sizes and longer training schedules than BERT, further strengthening the quality of learned representations. The model was pre-trained on approximately 20 GB of Vietnamese text, consisting of 1 GB from Vietnamese Wikipedia and 19 GB from Vietnamese news articles.

References

Abdelrazek, A., Eid, Y., Gawish, E., Medhat, W., & Hassan, A. (2023). Topic modeling algorithms and applications: A survey. *Information Systems, 112*, Article 102131.

Abuzayed, A., & Al-Khalifa, H. (2021). BERT for Arabic topic modeling: An experimental study on BERTopic technique. *Procedia Computer Science, 189*, 191–194.

Adel, H., Dahou, A., Mabrouk, A., Abd Elaziz, M., Kayed, M., El-Henawy, I. M., … Amin Ali, A. (2022). Improving crisis events detection using distilbert with hunger games search algorithm. *Mathematics, 10*(3), 447.

Alaei, A. R., Becken, S., & Stantic, B. (2017). Sentiment analysis in tourism: Capitalizing on big data. *Journal of Travel Research, 58*(2), 175–191.

Alcoforado, A., Ferraz, T. P., Gerber, R., Bustos, E., Oliveira, A. S., Veloso, B., Siqueira, F. L., & Helena, A. (2022). ZeroBERTo: Leveraging zero-shot text classification by topic modeling. *Lecture Notes in Computer Science*, 125–136.

Atandoh, P., Zhang, F., Adu-Gyamfi, D., Atandoh, P. H., & Nuhoho, R. E. (2023). Integrated deep learning paradigm for document-based sentiment analysis. *Journal of King Saud University - Computer and Information Sciences, 35*(7), Article 101578.

Azhar, A. N., & Khodra, M. L. (2021). Fine-tuning pretrained multilingual BERT model for Indonesian aspect-based sentiment analysis. *arXiv*.

Blei, D., Ng, A., & Jordan, M. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research, 3*, 993–1022.

Carrasco, P., & Dias, S. (2024). Enhancing restaurant management through aspect-based sentiment analysis and NLP techniques. *Procedia Computer Science, 237*, 129–137.

Catelli, R., Pelosi, S., & Esposito, M. (2022). Lexicon-based vs. BERT-based sentiment analysis: A comparative study in Italian. *Electronics, 11*, Article 374.

Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv (Cornell University)*.

Ding, K., Niu, Y., & Choo, W. C. (2023). The evolution of airbnb research: A systematic literature review using structural topic modeling. *Heliyon, 9*(6), Article e17090.

Egger, R., & Yu, J. (2022). A topic modeling comparison between LDA, NMF, Top2Vec, and BERTopic to demystify Twitter posts. *Frontiers in Sociology, 7*.

George, L., & Sumathy, P. (2023). *An integrated clustering and BERT framework for improved topic modeling*.

Gokcimen, T., & Das, B. (2024). Exploring climate change discourse on social media and blogs using a topic modeling analysis. *Heliyon, 10*(11), Article e32464.

Grootendorst, M. (2022). *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. ArXiv:2203.05794 [Cs].

Gulati, K., Kumar, S. S., Boddu, R. S. K., Sarvakar, K., Sharma, D. K., & Nomani, M. (2022). Comparative analysis of machine learning-based classification models using sentiment classification of tweets related to COVID-19 pandemic. *Materials Today: Proceedings, 51*, 38–41.

Gunadi, J., & Setiawan, B. (2024). Application of natural language processing (NLP) for multilingual tourism: Google translate for effectiveness communication. *Asian Journal of Language, Literature and Culture Studies, 7*(3), 559–572.

Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the knowledge in a neural network*. arXiv preprint arXiv:1503.02531.

Hotel Tech Report. (2026). *75+ hospitality statistics you should know.* https://hoteltechreport.com/news/hospitality-statistics.

Jayathilaka, R., Dharmasena, T., Rezahi, N., & Haththotuwegama, S. (2020). The impact of online reviews on inbound travellers' decision making. *Quality and Quantity*.

Jelodar, H., Wang, Y., Yuan, C., Feng, X., Jiang, X., Li, Y., & Zhao, L. (2018). Latent dirichlet allocation (LDA) and topic modeling: Models, applications, a survey. *Multimedia Tools and Applications, 78*(11), 15169–15211.

Jurafsky, D., & Martin, J. H. (2024). *Speech and language processing* (3rd ed.) draft).

Kausar, S., Huahu, X., Ahmad, W., & Shabir, M. Y. (2019). A sentiment polarity categorization technique for online product reviews. *IEEE Access, 8*, 3594–3605.

Kim, K., Park, O., Yun, S., & Yun, H. (2017). What makes tourists feel negatively about tourism destinations? Application of hybrid text mining methodology to smart destination management. *Technological Forecasting and Social Change, 123*, 362–369.

Kukreti, A., Kaur, A., Altaf, O., & Adil, M. (2024). Travel industry: Comparison of travel review for ease in making travel decision. *SSRN Electronic Journal*.

Kumar, A., & Sachdeva, N. (2021). A bi-GRU with attention and CapsNet hybrid model for cyberbullying detection on social media. *World Wide Web, 25*(4), 1537–1550.

Le, N. B. V., & Huh, J. H. (2021). Applying sentiment product reviews and visualization for BI systems in Vietnamese E-commerce website: Focusing on Vietnamese context. *Electronics, 10*(20), 2481.

Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., … Stoyanov, V. (2019). *Roberta: A robustly optimized bert pretraining approach*. arXiv preprint arXiv:1907.11692.

Luo, J. M., Vu, H. Q., Li, G., & Law, R. (2021). Understanding service attributes of robot hotels: A sentiment analysis of customer online reviews. *International Journal of Hospitality Management, 98*, Article 103032.

Mahdikhani, M., & Meena, P. (2024). Metaverse applications and supply chain innovation: Insights from text mining. *Journal of Innovation & Knowledge, 9*(4), 100591–100591.

Mahendhiran, P., & Kannimuthu, S. (2018). Deep learning techniques for polarity classification in multimodal sentiment analysis. *International Journal of Information Technology and Decision Making, 17*(3), 883–910.

Marcolin, C. B., Becker, J. L., Wild, F., Behr, A., & Schiavi, G. (2021). Listening to the voice of the guest: A framework to improve decision-making processes with text data. *International Journal of Hospitality Management, 94*, Article 102853.

Mehraliyev, F., Chan, I. C. C., & Kirilenko, A. P. (2022). Sentiment analysis in hospitality and tourism: A thematic and methodological review. *International Journal of Contemporary Hospitality Management, 34*(1), 46–77.

Ngo, T. M., Ngo, B. H., & Valerievich, S. V. (2024). Fine-tuned PhoBERT for sentiment analysis of Vietnamese phone reviews. *CTU Journal of Innovation and*

Sustainable Development, 16(Special issue: ISDS), 52–57.
Ngo, V. S., Nguyen, T. M. N., Hoang, T. H., Nguyen, H. L., & Vo, V. M. N. (2022). An improvement of PhoBERT to enhance the Vietnamese language understanding capability of a hotel information chatbot. *Hue University Journal of Science: Techniques and Technology, 131*(2A), 5–20.
Nguyen, H. T. T., Huong, T. P., Tram, A. L. T., & Tran, T. V. (2023). Exploring customer feedback on their hotel experiences in Vietnam. *International Journal of E-Entrepreneurship and Innovation, 13*(1), 1–17.
Nguyen, D. Q., & Nguyen, A. T. (2020). *PhoBERT: Pre-trained language models for Vietnamese*. arXiv preprint arXiv:2003.00744.
Nguyen, H. T. T., & Nguyen, T. X. (2023). Understanding customer experience with Vietnamese hotels by analyzing online reviews. *Humanities and Social Sciences Communications, 10*(1), 1–13.
Nguyen, D. Q., Nguyen, D. Q., Vu, T., Dras, M., & Johnson, M. (2017). *A fast and accurate Vietnamese word segmenter*. arXiv preprint arXiv:1709.06307.
O'Callaghan, D., Greene, D., Carthy, J., & Cunningham, P. (2015). An analysis of the coherence of descriptors in topic modeling. *Expert Systems with Applications, 42*(13), 5645–5657.
Rahman, H., Tariq, J., Masood, M. A., Subahi, A. F., Khalaf, O. I., & Alotaibi, Y. (2023). Multi-tier sentiment analysis of social media text using supervised machine learning. *Computers, Materials & Continua, 74*, 5527–5543.
Ruiz, C. A., & Fernández Avilés, G. A. (2023). Tourism destination events classifier based on artificial intelligence techniques. *Archivo Digital UPM*.
Saidi, F., Trabelsi, Z., & Thangaraj, E. (2022). A novel framework for semantic classification of cyber terrorist communities on Twitter. *Engineering Applications of Artificial Intelligence, 115*, Article 105271.
Sánchez-Franco, M. J., & Rey-Moreno, M. (2022). Do travelers' reviews depend on the destination? An analysis in coastal and urban peer-to-peer lodgings. *Psychology and Marketing, 39*(2), 441–459.
Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). *DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter*. arXiv preprint arXiv:1910.01108.
Sann, R., & Lai, P.-C. (2020). Understanding homophily of service failure within the hotel guest cycle: Applying NLP-aspect-based sentiment analysis to the hospitality industry. *International Journal of Hospitality Management, 91*, Article 102678.
Singgalen, Y. A. (2024). Sentiment classification of robot hotel content using NBC and SVM algorithm. *Journal of Computer System and Informatics, 5*(2), 442–453.
Tan, K. L., Lee, C. P., & Lim, K. M. (2023). A survey of sentiment analysis: Approaches, datasets, and future research. *Applied Sciences, 13*(7), 4550.
Tang, R., Zhao, S., Lee, W. S., Park, S., & Zhang, Y. (2024). Crisis response in tourism: Semantic networks and topic modeling in the hotel and aviation industries. *Sustainability, 16*(24), Article 11275.
Tenney, I., Das, D., & Pavlick, E. (2019). BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th annual meeting of the association for computational linguistics* (pp. 4593–4601).
Thin, D. V., Hao, D. N., & Nguyen, N. L. T. (2024). A study of Vietnamese sentiment classification with ensemble pre-trained language models. *Vietnam Journal of Computer Science, 11*(1), 137–165.
Tripadvisor. (2019). *Online reviews remain a trusted source of information when booking*. https://ir.tripadvisor.com/news-releases/news-release-details/online-reviews-remain-trusted-source-information-when-booking.
Trung-Thanh, H., Ho Van Nguyen, Su, H. Le, Thi-Kim-Hien, Le, & Thien, Le B. (2023). Analysis of online user sentiment and behavior in the tourism sector in Vietnam based on reviews and comments. *VNUHCM Journal of Economics, Business and Law, 7*(1), 4089–4103. https://doi.org/10.32508/stdjelm.v7i1.1187
Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., … Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems, 30*.
Vu, T., Nguyen, D. Q., Nguyen, D. Q., Dras, M., & Johnson, M. (2018). *VnCoreNLP: A Vietnamese natural language processing toolkit*. arXiv preprint arXiv:1801.01331.
Wettig, A., Gao, T., Zhong, Z., & Chen, D. (2023). Should you mask 15% in masked language modeling?. In *Proceedings of the 17th conference of the european chapter of the association for computational linguistics* (pp. 2985–3000).
Wu, S., Xu, Y., Wu, F., Yuan, Z., Huang, Y., & Li, X. (2019). Aspect-based sentiment analysis via fusing multiple sources of textual knowledge. *Knowledge-Based Systems, 183*, 1–18.